

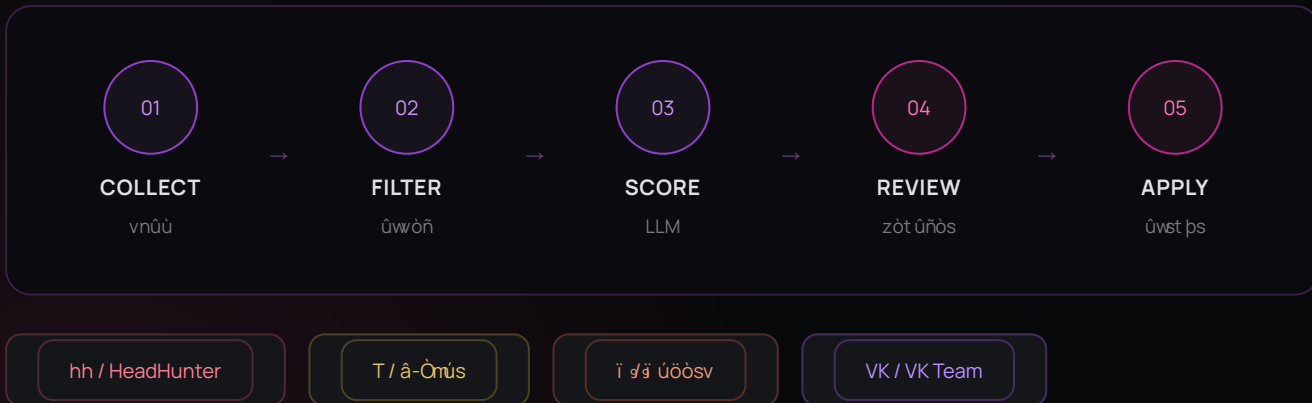
hh- agent

.SYSTEM DOSSIER

LOCAL-FIRST AUTOMATION / PRODUCTION ARCHITECTURE

á pvnòu ún6 öûsxu òúwrfp6

Öüÿpwòswùmj LLM-ûffòúsmjsúúwùút pùxòu° òúwst psp,
úmt 5 òmòu úvwtpnòpûümv ún6öüvwñsmñ production.



CODE SNAPSHOT

main @ 3d31520

UPDATED

14 òúw6nù62026

CONTROL

explicit approval

OBSERVABILITY

read only

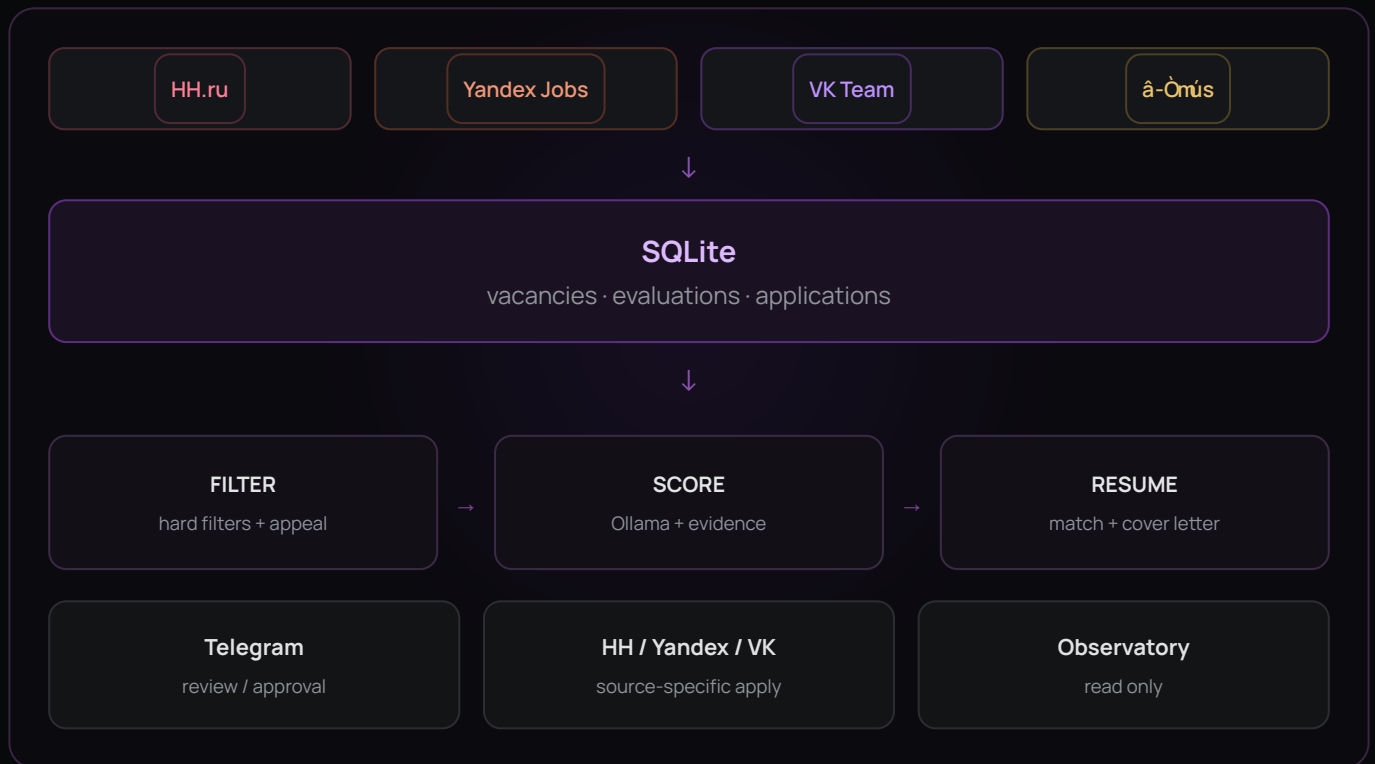
á ûöòùœúþò

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2. 2 vûsûxúúñûòñm6mûÿpwòsvxùm	12. Windows Scheduler þshared AgentLock
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HH Agent — t ùsmt tñ° q Windows-first modúwáúòpvs mñmñw . ý úa ùnpòmòwñkñmñvpppñsòvñsùt tñpñvñwñzùpsññ, ùwñmñ° ãmòwñzòñpòbùññòbùñöyöò6fi pò, ðmòwñvñùùù° u ðhard-filter reject ùmññùñmñLLM-mñòt t 6ffp5 , ðñ ùñt ù6òw vñwñxwñùpùññmñúx5 ãLLM-ùffòúsx, ðñùñpòmòwñmòp5 u ò, ðmñwññpñwñvacancy-aware vñùùñññùñpòwòt tñùòòñpvtñuñ, ùñsmñp° ãmòwñmòsñu òòñffpñpññTelegram pñwñmñmññt 6òwññt tñsñwñòñwñst psp, sñwñùñ° òñññùññòññùpñt ðñt tñpññmñwòt tt á nñù, ðñffòúsmñmòòffñbùpòòññt tñpññmñwòt 6pñsite-specific apply adapters ùmpñòòòò° ðñùñmñpñ° u sñùñwñùmñ, ðwññ° pñu òùòùpòòpñùwñùý òqñvñmñmñmñòñuñt ðñùñpñmñòwñùññu òùòwñtñùñññt mñffòúspñt pñmñpñùòffñbùpòpñmñmñwñùñmñsx.

Ùúúvù	âðsxfi òòûññîòòóùþò
Discovery	HH.ru, Yandex Jobs, VK Team, â-Ôñús
Filtering	hard filters + LLM appeal öt 6ãüüúú° ýreject + role/domain policy
Scoring	t úsnt ttú° qOllama, structured output, evidence guard, management policy
Resume	ñ° núûûüüý þt 6/ùòþ5 u ò, single-resume experiment þæwòt òu òwùþ6
Cover letter	òòúòèþùxòw 6ûüòôûñúòùòwùx5 ãñsmúnþ5 þþvüüt tpxòwåût tsûûûüöwñòùòôòùú° òý nsw
Approval	snoö° qáwst þæwùònxòw6ñúúóûûüt tpûñmòt tvsúóú: approved
Apply	üwòòt ttú° òæworkers öt 6þHH, Yandex þå/K; â-Ôñús— discovery only
Telegram	ùòòûuòúömfþþ, health/status, ùxzúüqpnüxvs, recovery þJURL → cover-letter flow
Resume Raise	smart supervisor, súwù° qáwsù° ñmòwê-H wût tsúûûüöþömbuúuxñùòuòùþãòñvüt mwúóûûüò° öum
Observatory	t úsnt túñ6read-only ümòt tLIVE / ANALYTICS / RESUME
Targeted Hunt	üòþvsæwzspñýüömpæòt òñnúwúúóúûûwmswnñsûuüüþþ
CI/CD	branch → PR → CI → main → CD → Octopus → production PC

2.0 vûsûxùûñúòññ6mùÿpwòswwùm



â úúûñ° qpipeline

```
check_hh_session
  ↓
hh_collect_optimized.py
  ↓
collect_careers.py
  ↓
process_vacancies.py
```

Apply òpñòwâwòòt ttúúûw pipeline:

```
background_apply.py
  ↓
apply_dispatcher.py
  ├── HH      → apply_worker.py
  ├── Yandex  → yandex_apply_worker.py
  └── VK       → vk_apply_worker.py
```

g wûûmpòòt òúpòsùùpúffpùpnt ttúú. Pipeline u úòòwâûnpùmtùpúffòùpññwttñsmùvppnòpñmùxvsmùwst psùñ. Apply supervisor zpwnòwâwòòt tsùûññúûûmpùòffòúúx5 úzòùòòtt Observatory úpzòòúúòúòùòòwpsùúòst 5 zòússòmùú° u wùt tsùúûmzvòùpò.

Vacancy

āmúpwaíúúunt ppûñmúx5 3msnúp5 pꞤpöòúwpzúúvvtqvwúúpsmġmúúpꞤòvsn6pꞤpöòúwpzúúvvt- [source +](#)
 external_id; hh_id 3wÿÿmúóúót 64egacy-vûñuòvwpuúvvp4H.

Evaluation

Úwùþ6áffòúúsxüýùnóòw6öt 6xöþwmðúòqúnŷüô6w6fwûüñ qscore þdecision, role/seniority/domain/
responsibility fit, gaps þred flags, ùòûuòúöñffþ6,xòòùòþûüñkíúúûò÷xüüüññöþwòt ttúò,ñ nùnúú qresume key þ
uòwòtmúú òmatching.

Application

ãumúpwüüt tpuñmwòt tvsuòuòffbuòpòpoperational lifecycle ûvst psm

á wmwv	á u° vt
notified	smúwúzsrmúúsnpmúmhTelegram, ùòfffbúp6òfi òáòw
approved	Üüt tpúñmwòt tēñúúánpùòffpt áwst ps
applying	worker únzmt ánnúmwxs
waiting_captcha	úxæúáxzúúqCAPTCHA / vntwæðwzòt úñòsm
applied	xvüöýáúüöwñòúæðú, pmüüt úóú: applied_at
manual_required	nñwúmvpsmòpúümvúúávwñúñpt mvtbæðwzòt úñòsm
apply_error	wòýúpzòvsn6úffpnsrmóúáúüöwñòúæðúúúqávwúññsp
skipped	Üüt tpúñmwòt tēüúüxvvtpt áñsmúvp5
company_blacklist	súu ünúp6ávwúññt òúmñblacklist flow

Y vuuññûqsafety-ûùpúffpû:ý nsw approved x/pt túòòùùòò° öxfi ööü.LM decision þáþúþmzñwöññùò
 ûùt tpûñmòt tvsùòùmpuòffþúpò. Evaluation üùvt òòwñùùùvñw6ñùùòsvñwù þþvñwùþòq,ñùòññùù° u ãsù° w u
 veto-gate.

4. Discovery: ùwvñzúpsmôûSQLite

4.1 HH.ru

Production collector pnúxvsnòw6zòùòp: `hh_collect_optimized.py`. Ý úámvzpwñmpûñ q: `hh_collect.py`, áú vñyùnú6òwòòûbrowser/session logic pwatchdog.

âòsxfi n6vñmòdp6:

- persistent Playwright profile `browser-profile`;
- ùùññòùsmHH-vònvppáòùòöñnùùu;
- ùòùvùunt ttú° òáòsùuòùömfppáñnùmmw ññ5 w6áòùñ° up;
- fallback target-role search pnúxvsnòw6vñt tsùòvt páòsùuòùömfppáòùòvñwñzúú;
- target-role queries ùùpùùpvpppùx5 w6;
- úmfallback SERP úzòñpöùùáòffòt òñ° òáitle úwòpññ5 w6áòùáñsù° vñ6smùwñzsp;
- xòöñyùñmòùù° òññsmùvpáñvñ5 w6ññSQLite òmòáùùpáùvt òòx5 fi òu pññpvnùppábrowser process;
- child collector sùùwñt pùxòw6watchdog, sùwñù° qñyùñvñòùáñvñmùññpññpñvffòòòùòññáùùffòvññ.

Òmpûñ° òpúnzòùp6:

```
HH_RECOMMENDATION_PAGES=3
HH_FALLBACK_MIN_NEW_FROM_RECOMMENDATIONS=10
HH_DELAY_BETWEEN_VACANCIES=7
HH_COLLECT_NAVIGATION_TIMEOUT_MS=30000
HH_COLLECT_WATCHDOG_SECONDS=120
```

`hh_collect_optimized.py` áòùùùt úpwòt ttùùxuòùtffñòwvñmù° òádefault delays uòöxñwñmùpffmupápnùùvñmupáòù 8pá0ñòsúùöñùùñòwñòùùù,òvt páùpáòáòùòùùùòòòòù° environment variables.

Qvt pñHH-vònvppáùùwñyñt m pipeline úòáùùòuòù6òwáòùvùunt ttú° qñnùùáñ° zú° u áùùpvsùu: HH collection ùùùùxvsnòw6,áùùt tpùññwòt ttáùùt xzmòwáùùòòxùùòöòùùpò,ññùùùùùmvpññ° òpñwñzúpspáòñt ttùòqffñ6áñnùmmùwsm ùùùòùt çñ5 w6.

4.2 Yandex Jobs

`sources/yandex.py` ññ° p° ññòw6zòùòp: `collect_careers.py`. Ávñwñzúpsppùt pùññmùáñwòùxqpýcareer collectors. Apply ññ° ùùt ú6òw `yandex_apply_worker.py` áùùt tsùáòt 6° `approved` pñáùùt tsùáùùp: `YANDEX_APPLY_LIVE=true`.

4.3 VK Team

`sources/vk.py` áññpùñòwññsmùvpññVK Team. Worker xuòòwppñùùt ú6vññcontact/about/social fields, pñùxòñvñt ññ° nùmùùùòáòp5 uò,áñnùmmw ññvññconsent, ùòmùpzoùùùöñvññxúx5 CAPTCHA pñmñùùpùññvññxvùòffùx5 ùwùññsx,áñffpñsxá pñùstructural form change.

4.4 â-Òmús

`sources/tbank.py` — discovery-only source. Ý úáñzòwññstatic pagination pñdynamic Playwright discovery, xuòòw ùñyùòpññlazy-loaded catalog, òòòxùt pffpùxòwññsmùvpññáùùùUID, ùwñmñ° ññòwññinactive/closed pñùùpuòù6òw

management/product/project title policy. Transient HTTP errors will be handled with a bounded backoff.

â-Ômúsámuòòòúúûúòúûmúffúxwpppúxòw6ñautomatic apply: úwòòt túúôûs `tbank_apply_worker.py` 9òqzmvúòw

4.5 Fault isolation career-pvwûzúpsûñ

collect_careers.py

5. Evaluation pipeline

âòsxfi pqqüù6öûsûnùmnûwsp:

```

Hard filters
↓
LLM appeal для appealable reject
↓
Structured evaluation (Ollama)
↓
Evidence guard
↓
Deterministic management policy
↓
Resume matcher / selector
↓
Persistence + Telegram eligibility

```

5.1 Hard filters

[illegible]

5.2 LLM-muòt t 6ffp6hard reject

Yòsmæ qreject úsúúzmòt óú. ð t 6appealable cases **HardFilterAppealReviewer** úúúúpwð LLM úúúúòòt pwtúúò
6ñt 6òw6ð ðhard reject false negative.

- verdict: `confirm_reject` or `override_reject`;
- confidence: `0.0..1.0`;
- default threshold or `override`: `0.75`;
- unprompted structured response;
- prompt `tuúwúònt tuúwúònxòwèxpwr ñmwtuònvnuòmnuú° òàuxvvpò/múct pqyspò` titles, C-level, Director/Head p
ùònt tu° òân6pmúúvvp, úòàùùbòxu° ññ6smúöbömwxúú° w

Úvwûùpzòvnbòreject uôoxwüòuòûffòùbñmwtv6ûwòt tú° u ð backfill flow.

5.3 Structured scoring

Ômpũĩmũyũ uũxt m

```
role_match          * 0.35 +
seniority_match     * 0.20 +
domain_match        * 0.15 +
responsibility_match * 0.30
```

Evaluation pōōwzòbòpā ūsrt tú° qōllama pstructured schema. Evidence guard ūūhōūō6ōwzwxñ° ñhōōūūpūnòw6 ūmñmñv5 pñūōwñōōōū° òy msw ãūūy pt 6. Management policy pñōū ãūpu óū6ōwōwōū pūpūññū° ò smñtòū° òñūññpt mññhōūy LLM ūōpxt tñwñw

5.4 LLM resilience

Qvt pGPU pm6wpt pDollama ñòuòóúúôöüvwxùó, scoring/appeal úòàùòèñnfí n5 w6ñđ úóé° qreject. Ørsmúvp6
úvwrtw6pending þuúçèwñ vttûnùmûwmûrt öðx5 fi þu àùùýûôûu. Ðt þúú° qþðùùñ° q process_vacancies.py
úòàùòmúpzòúsñfi þu wall-clock timeout; timeouts úvwrt w6ámxùùñúòsúúsùòwú ýñúòfflþýáùòònnfbq.

6. ¿Qué es un sistema de gestión de la calidad?

Resume selection üüûpvÿüöpwöü site adapter. Ø° númúú° qđ úsmt tú° qđ PDF ñmt pöpxów6zòuòp: [app/application_assets.py](#) ðáumvvunwüöñmòw6smss submission source of truth.

á ûüùûñûöp wòt ttúûòüpvttu û:

- vûpömtw6mûöšmûsùöwúx5 ǵĩrsmvnp5 ;
- vñ6p° ħñòwvñt tsñúòt òħmúwñ° òmûöwñòòöõúú° òǵ msw° mûúy pt 6wñòññmúp6u pñút p;
- úòvñöòòöwñ placeholders;
- úòǵñ° öxu° ħñòwñü° wñwòšpt pöñvñpöúp6;
- vñÿñmú6öw6ǵ Evaluation/Application context bñmòu ðvñút tpxöw6wñwñ'ñu.

Vacancy URL → cover letter зòуòрTelegram

Zit ttpũĩmòt ttu ũcœwũùpvtt mttv° t sxũmñĩsmũvp5 . Flow smũũpppũxòwJRL, ũũũòòò 6òwH/Yandex/VK, ũũ ãĩpuũcœũũvw pũvũt tpxòwœœvũỹũmũõũũòEvaluation letter, pũnzòũũt xzmòwvacancy content pœòòũpũxòw ũũũòũpvttu ũzòòòpũt ũsrst ttũũòũòp5 uò, preferences pœOllama. p t 6ũòpprònvũ° ħpublic HTTP(S) vntqũũgeneric fetch path ũcũmũpzpñmòwlocalhost/private/service networks, ũòvnũũömwũ° òũũũw pũòòũũũũt pũxòu° ò redirects.

- [illegible]

DOM/regression tests, 5 delayed post-apply UI, 6 submit targeting guarded retry.

9. Telegram – control plane

Production entry point: `telegram_bot_entry.py`

Ùúuúôm	Zũñòòòúpò
/health	system/Ollama/runtime/queue health
/status	ý ûúúñ° òûvû6úp6þbreakdown approved queue
/run	pnûxvsrnòwpipeline vòqzm
/new	unresolved/manual-required snùwûzsp+úúñ° òòòúu òúõnffpp
/stats	vvnþvvpvsmApplication statuses

/new မှီဝဲတံ ယ်ပုၤခဲၤတံၤ

[illegible]

Recovery pmanual_required

/new ၁/únznt mñpññfi nòw manual_required, ၃nòv unresolved notified, ၄úvt òzòòúñmñt 6òw òòqvnñpòt tuúúúñ òòòòúúòñffpp. Best-effort notification ú manual_required ၁/úòòòúñmñp5, súuúñp5, ၄úvpzbx, Application ID ၃súúúsxúñsù° wb6ñmñp.

Telegram Watchdog

Ý wòòt tũn6scheduled task HH Agent - Telegram Watchdog ãũòũñòũ6òw6Telegram ùmpñũpũxw.Š ũũwvt òòpñmòw runtime heartbeat, PID ãý mwpzòvspqprocess command line. Default stale threshold — 180 vòsxüö.ž ũũpñpũvffũb heartbeat ãt ãpũvzòpũxñffũb process watchdog ũòũòpñũxvsmòw scheduled task, ũũòòñmũp wòt tũũũñpññ6stale telegram bot entry.py process.

- UI úòpñxvsmòwworkers þáðáúòú6òwbusiness state.

Ývsnú° wttt úsnt ttúú: `http://127.0.0.1:8765`.

12. Windows Scheduler þshared AgentLock

Scheduled task	ænnvúwn/trigger	Entry point	Ývúnoúúúvþ
HH Agent - Pipeline	smæð° òðznm	<code>background_pipeline.py</code>	HH session guard, runtime heartbeat
HH Agent - Apply	smæð° òð0 upúxw	<code>background_apply.py</code>	zþwòwútt tsúapproved queue
HH Agent - Resume Raise	smæð° òð5 upúxw	<code>background_resume_raise.py</code>	heavy browser only near <code>next_due_at</code>
HH Agent - Telegram	üüþðogon	<code>telegram_bot_entry.py</code>	pythonw, restart policy, IgnoreNew
HH Agent - Telegram Watchdog	smæð°x5 upúxw	<code>telegram_watchdog.py</code>	stale heartbeat / PID recovery
HH Agent - Dashboard	üüþðogon	<code>dashboard</code>	local read-only UI

Pipeline, Apply þResume Raise þvüút tpx5 wúñfi þqAgentLock. g vúúðáüþñútt 6òwúúúy t þswx5 fi þu þbrowser jobs úmúwntúúmt t òt ttúþáúðffwttòúxóðúxóæzòòþúúúy þt þ, UI þt þnetwork state.

Production deployment þvüút tpxòwúwæðAgentLock, zwúñ° Git update úðáüüþvñúðpt áúvòúððrúððþþñúq browser operation.

13. Runtime state þt úðþ

Runtime JSON:

```
data/runtime/pipeline.json
data/runtime/apply.json
data/runtime/resume_raise.json
data/runtime/telegram.json
data/runtime/telegram_watchdog.json
```

Ývúúñú° òðt úðþ:

```
logs/pipeline_supervisor.log
logs/collector.log
logs/careers_collector.log
logs/processor.log
logs/apply_dispatcher.log
logs/apply_supervisor.log
logs/apply_worker.log
logs/apply_worker_attention.log
logs/yandex_apply_worker.log
logs/yandex_apply_worker_attention.log
logs/vk_apply_worker.log
logs/vk_apply_worker_attention.log
logs/telegram.log
logs/telegram_watchdog.log
logs/resume_raise_supervisor.log
logs/resume_raise_worker.log
```

Browser profiles, `.env`, SQLite DB, runtime JSON logs – it's not too operational data but it's still useful.

14. CI/CD for production deployment

It's a bit more complex than the PR-first flow:

```
feature branch
  ↓
pull request
  ↓
GitHub Actions CI
  ↓
merge to main
  ↓
CI on main
  ↓
GitHub Actions CD
  ↓
Octopus Deploy
  ↓
HH-Agent-PC
  ↓
C:\hh-agent (fast-forward only)
```

CI

Windows / Python 3.12 pipeline is a bit more complex:

- `git diff --check`;
- Python compileall;
- syntax validation `deploy/octopus_deploy.ps1`;
- xvfb + Chromium for DOM regression tests;

- core `unittest` suite.

CD / Octopus

Zúvt òxvüòffíúúòûCl úm **main** CD vùpöñõwrelease/deployment ñsOctopus Deploy project **HH Agent** òt 6 environment **Production**. Target — Windows PC **HH-Agent-PC** zòùòpPolling Tentacle.

```
deploy/octopus deploy.ps1:
```

- ðððmüxvsnòw Pipeline/Apply/Resume Raise;
- cæðwöû2zmvûñûvnñûnûcæðuþ6 AgentLock;
- xööùocþñròw AgentLock ãövttGit update + validation window;
- abort üüpãtracked local changes;
- harmless untracked files ûvwmtt 6òwámuònvò;
- zpvw qcheckout uûcæwñòúúxwtv feature branch úm main ;
- üüpúpunòwút ttsûfast-forward s; origin/main ;
- ñr üüt ú6òw Python sanity checks;
- üöùòpmüxvsrnòw Telegram wút ttsûüüpãuòt ôñnúw° ýppuóúouþ6ý;
- üüp post-update validation failure ñûpnñunfi nòw checkout sæüòö° öxfi òuxSHA

Octopus commit SHA, production deployment UUID, and Git revision.

GitHub CD úpúnzmówzwúpmiúvúmd deployment ñ Octopus vúpóm. úpúnt tú qtarget-side
 status úxóúvúúwúòvtn Octopus: target u úcòwúmwúxóúvúv òúúúzmúp6GitHub job.

15. Safety invariants

gwbäüñm̃p̃t m̃zpwñ5 w6zmvñ5 mü̃p̃wòswù° p̃öû̃t aë° mü̃òùòc̃ñm̃tùòý ñswùùp̃úo:

1. **Source separation.** Worker ùòèõùt cøwæpmpòmwtúizòùòöttøùxóuôssource.
2. **User approval is authoritative.** `approved` æipúnzrnòwñíúòòámnpùòffbüpòðáùìt tþûñmòt 6.
3. **External live switch.** Yandex/VK final submit wònòxòwæùîwñòvwvñx5 fi bqç `*_APPLY_LIVE=true`.
4. **No blind retry after submit.** æ'òüöùîpúnzú° qámòpxt ttvmæùòámüpñüòðpwæsmùñkñwùùúqámwùùññsò.
5. **Cover-letter confirmation ≠ application confirmation.** æ t 6æIH 0wùámnpú° òðûsmprnòt ttvñim
6. **Local resume is the submission source of truth.** Site adapter ùòámüpöxu° ñnòwøùxóuqresum asset.
7. **Evaluation history remains auditable.** ævwmùp6ámòæönt 6òw6ámòþæöüöñnvñim
8. **AgentLock ün6pnòt óúæ't 6æúýy t þswx5 fi þýðbrowser jobs þproduction deployment.**
9. `/new` áóðáòùòüpv° ñnòwæùòffbüp6. æ'úññvvwnñññt þñnòwunresolved þæünnññt 6òwáùñ° òsmùwúzsp.
10. **â-Ônúæùòæautomatic-apply source,** æùismúòæùpömúdedicated adapter.
11. **Observatory read-only.** Dashboard ùòæuúöcøwæuòú6wtbusiness state.
12. `main` æuòú6òw6æòùòppbranch + PR. Production üùt xznòwüùt tsùáüüüffböffbqçCI süò.

13. Production update — fast-forward only. Tracked t úsnt ttú° òφpu òúòúp6áúò&úpzwúœ&5 w6&òút ùòu.

16. Security þφpñònvú° ò&úònúpzòúp6

Ý nt nvwt	âòsxfi òò&úvnú6úpò
Telegram access control	backlog: public mode, úxœ&ú&allow-list
Dashboard exposure	wút ttsú&localhost; úò&úxnt þsú&ñwt&ú&nú&xœ&pp-pm& ú&ú&ñ/ú&ò&ú&ú&nt ttú° ý&ó&ú&ú° ý
Secrets	<code>.env</code> , tokens, browser sessions þ& þ&ú° ò&form values ú&ò&sú&u þ&w&w&6
T-Bank apply	ú&w&x&w&ñ&x&ò&w&ú&ú&ð&pp&ñ&ú&x
HH UI	u ò&ú&6&ò&w&6&zn&nv&ú; DOM regressions þ&ñfi þ&ñfi n&5 w&ú&mt 5 ò&ñ&ñ&ff&þ&ò&v&6&ñ&m&ú&pn&ú&w ,ú&ú&structural changes u ú&ò&x&w ú&ú&w&ò&ò&ú&ñ&ñ&wt&ú&manual flow
Ambiguous external submit	manual review ú&ú&ò&ú&ú&z&vp&ò&t ttú&ò&ò&ñ&w&ú&u m&þ&z&ò&nsú&ò&ú&ú&ñ&w&ú&ò&ú&þ&6
Runtime JSON	operational snapshot, m&ú&ò&ú&ú&t ú&n&6&event-sourcing þ&v&w&ú&þ&6

17. Operational runbook

Zú&ú&ñ&ò&ú&þ&wt&production checkout

```
cd C:\hh-agent
& "C:\Program Files\Git\cmd\git.exe" rev-parse HEAD
& "C:\Program Files\Git\cmd\git.exe" rev-parse origin/main
& "C:\Program Files\Git\cmd\git.exe" status --short
```

HEAD þ& origin/main ò&út œ&ú° &ú&ú&ñ&ò&ñ&wt& tracked working tree ò&út œ&ú&ú° wt&z&pv&w u.

Ö&ò&ú&u ú&w&ò&wt&ú&ú&ú&nt ò&u&x&pipeline

```
data/runtime/pipeline.json
logs/pipeline_supervisor.log
logs/collector.log
logs/careers_collector.log
logs/processor.log
```


19.09.2026 PDF 04.09.2026

á wñ° qPDF nùt tiffbàúòûwñmœt production. Øwòsxfi x5 ãñòùvp5 ãûñmñt òú° ððvüñmñt òú° :

- Observatory LIVE / ANALYTICS / RESUME þóúread-only núþwòsæúm
- dashboard telemetry öt 6hard filters þsingle-resume experiment;
- LLM appeal öt 6æüüü° ýhard-filter reject þþvñùþzòvþqþbackfill;
- ùmvffþùòúúñ6üüt þwþsmæsenior IT leadership titles;
- `hh_collect_optimized.py`, ðmúúþqSERP gate þáúññ° òáúññþt mHH collection;
- heartbeat pipeline ñûññòu 6ðt þúúúúcollect/process þæú6wþòáúñfi òòúæwall-clock cap vþòúúúññúúþprocessing batch;
- ýúúúññ6: `/new`, æúwúññ6ñüt tffþáúòñt úspúðw `/health` þðúúxþòþTelegram commands;
- úwòòt tú° qTelegram Watchdog vñññúumþþzòvþpu restart þññþvffþòúññwñ
- smart Resume Raise schedule: wake-up smæ° òðæþúúxwñòþáúñvññ6úúúúþmüxvsmChromium;
- production CI/CD zòùòþGitHub Actions + Octopus Deploy + AgentLock;
- commit SHA ñOctopus deployment;
- hardened production deploy: fast-forward only, protection tracked local changes, rollback validation;
- ñæwñt tú° qHH instant-apply flow, òòòæúúúúñññðþwòt túúúðáúñþst ñð° ññòw6áúwòòt túúúáúñvt òáúþ5 uò;
- üúúññòúþþæúúúúòwúúúþtextarea/submit öt 6post-apply letter;
- guarded retry/verification vüüüñññðþwòt túúúúþáúwòòt tú° q: `manual_required`, òvt þáúñst þðòvññt, mþvñtu úáò üúðwññòúæðúú;
- regression tests öt 6þobserved HH DOM variants;
- ñæwñt túññ6æmúwñsource routing þsafety invariants.

20. Φ ûsxuòúw ì6öûu

- `README.md` – Summary of the project
- `dashboard/README.md` – Observatory dashboard read-only view;
- `doc/ci_cd.md` – CI/CD for production deployment;
- `doc/single_resume_experiment.md` – single-resume experiment;
- `doc/targeted_hunt.md` – Targeted Hunt;
- `doc/hh_apply_success_detection.md` – HH apply success detection;
- `LICENSE` – Project license.

HH AGENT / rudenko.one

Local-first automation · explicit user approval · observable production · safe external actions.